

# Statistical Word Segmentation: Emergent Structure in a Next-Character RNN

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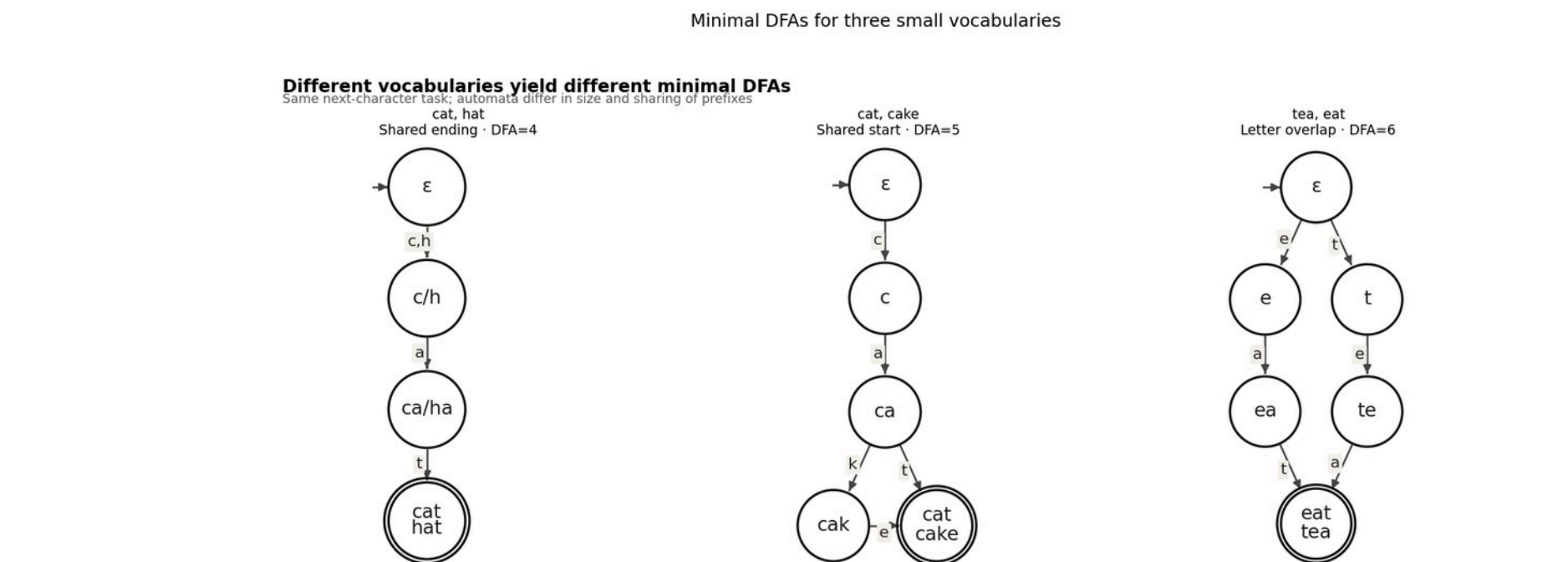
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## Abstract

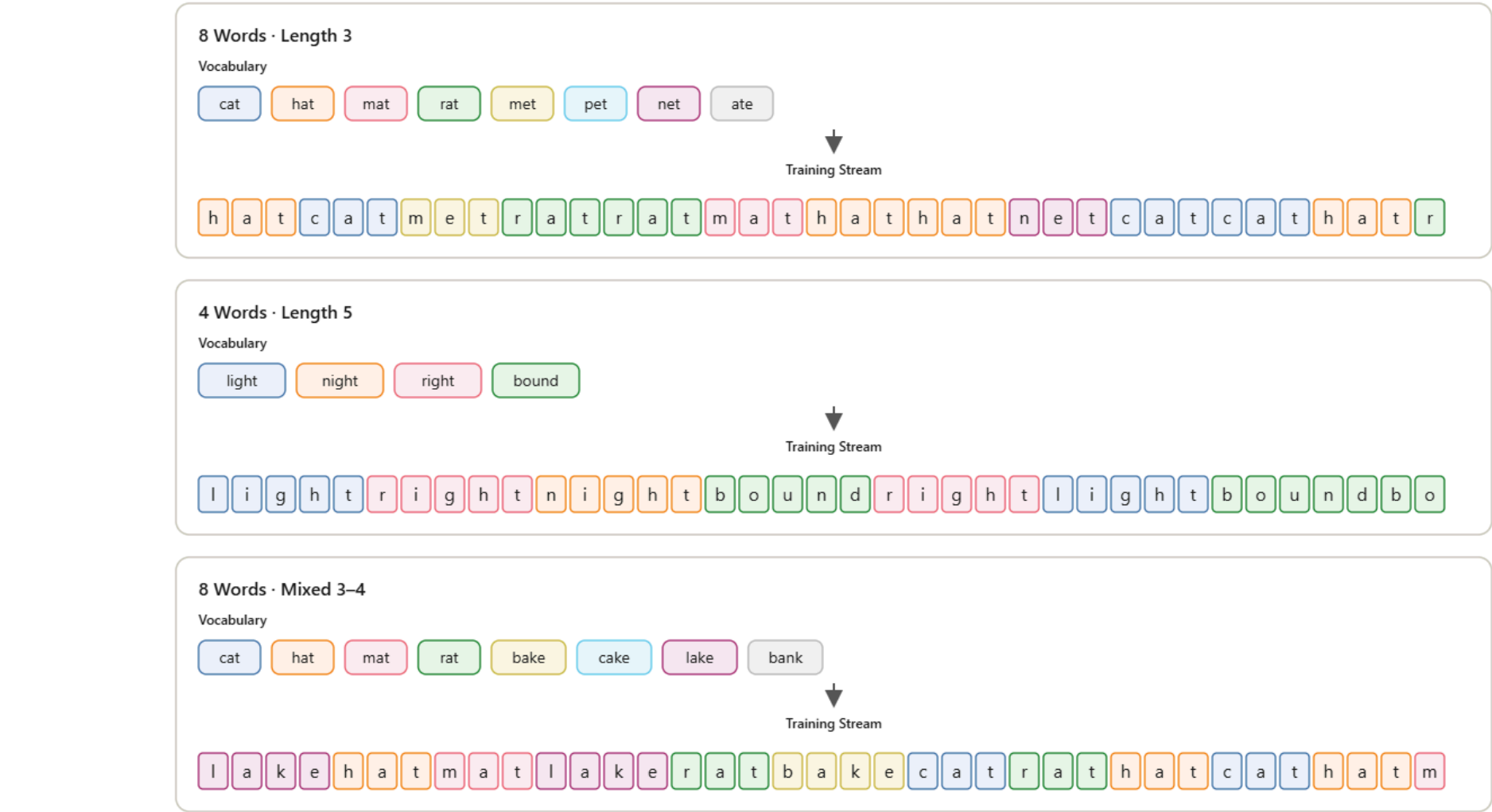
Eight-month-old infants can segment continuous speech by tracking transitional probabilities (Saffran, Aslin & Newport, 1996). We ask whether a vanilla Elman RNN trained only on next-character prediction develops internal representations aligned with word structure. After learning, the network generates legal vocabulary items, and its hidden states become a continuous embedding of the vocabulary's minimal DFA. On an eight-word -ate/-at demo, DFA state explains  $\eta^2 \approx 0.88$  of condensed hidden variance; word identity is much weaker ( $\eta^2 \approx 0.06$ ). Fifty mixed-length English vocab runs ( $H=100$ ) show that dimensionality, training cost, and readouts track minimized DFA size. Weight matrices become letter-columnar in  $W_{ah}$  and locally clumped in  $W_{hh}$ , most clearly for small automata.

Fig 1 • Minimal DFAs



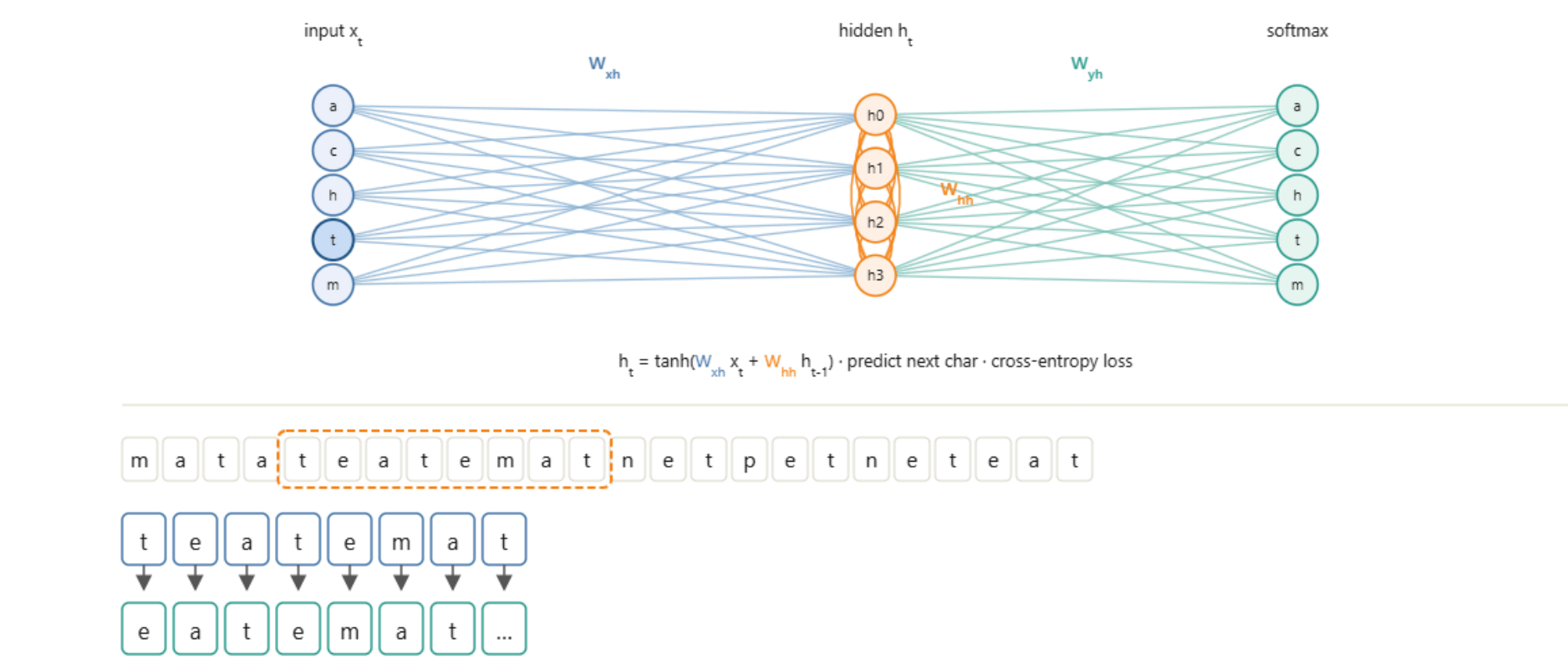
Same next-character task; automata differ in size and prefix sharing.

## Experiment • Vocabulary conditions



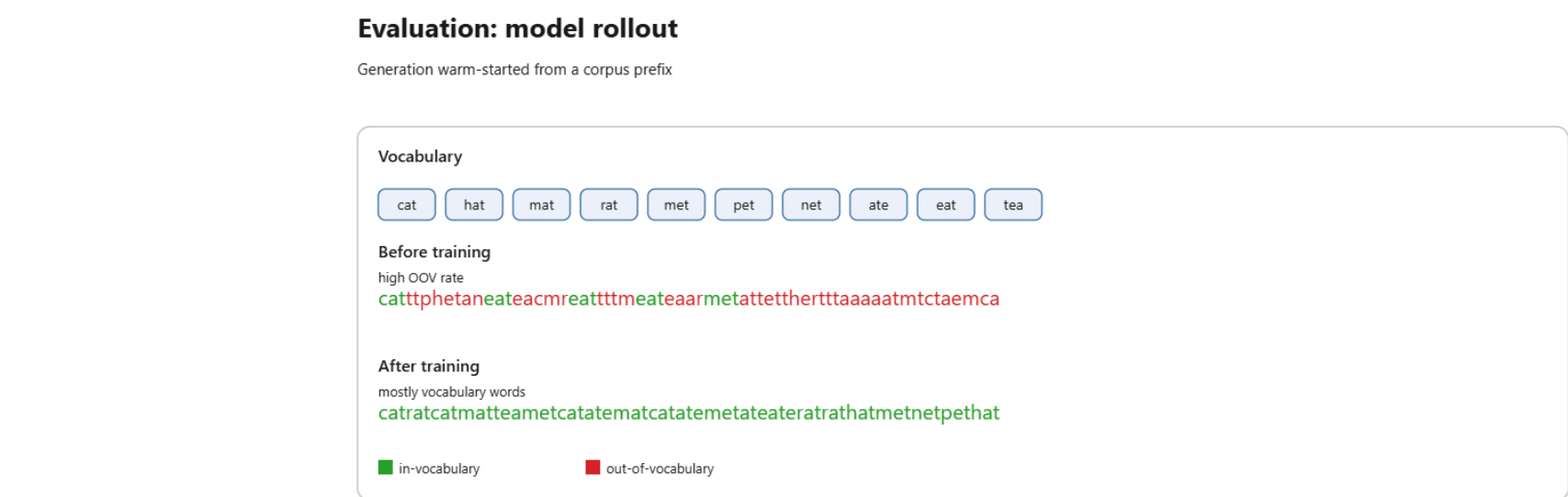
Same setup, different #words and lengths: 8x3, 4x5, and mixed 3-4. Colors mark word identity in the unsegmented stream.

## Experiment • Training the RNN



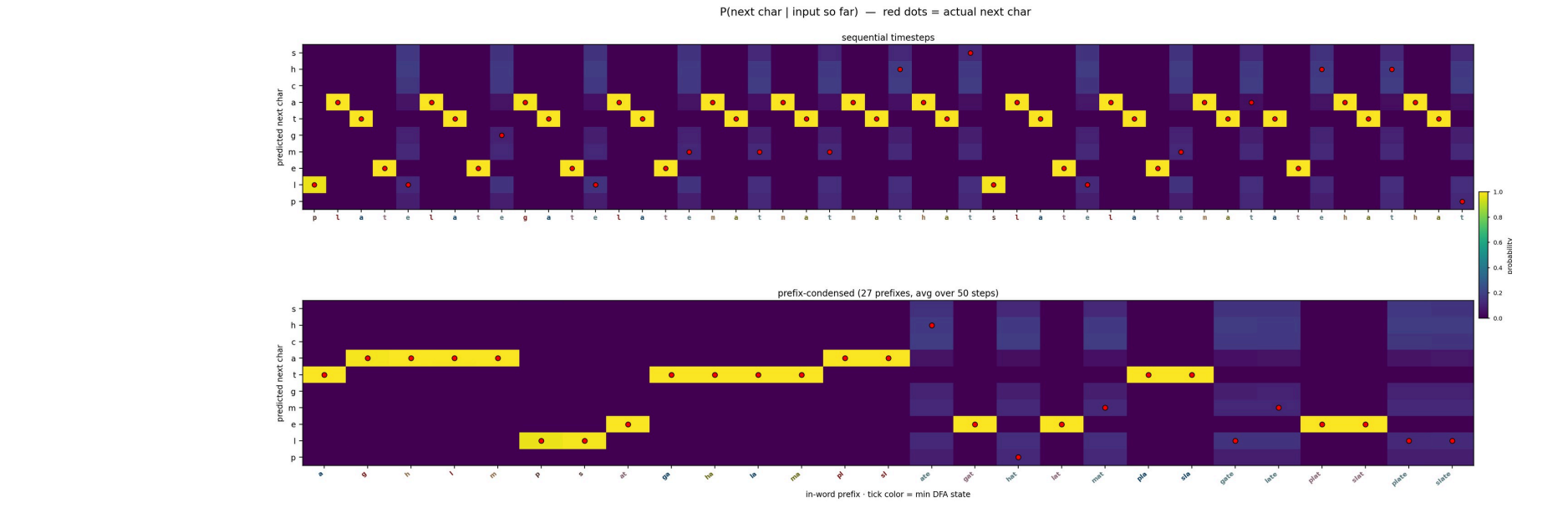
Character-level Elman RNN. Input  $x_t$  predicts the next character via  $h_t = \tanh(W_{ah}x_t + W_{hh}h_{t-1})$ ; cross-entropy loss.

## Experiment • Evaluation



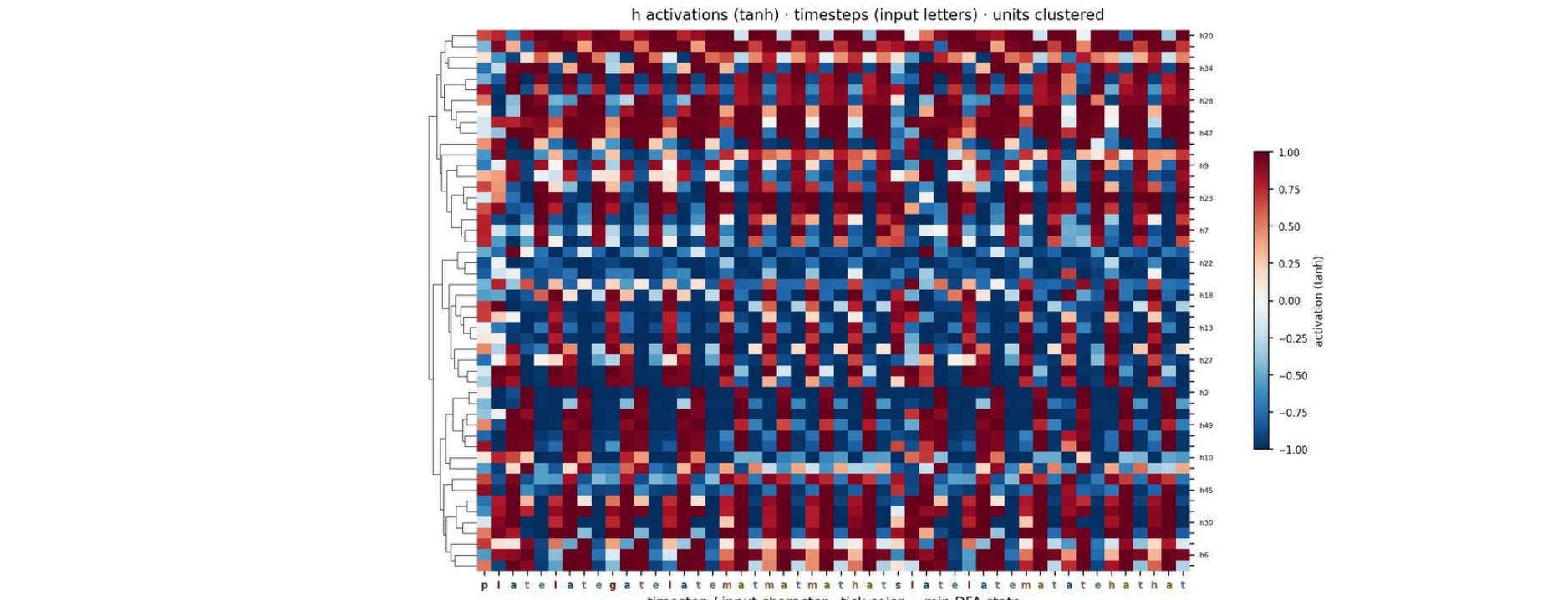
Closed-loop rollout from a corpus prefix. Green = in-vocabulary words; red = OOV.

Fig 4 • Next-character probabilities



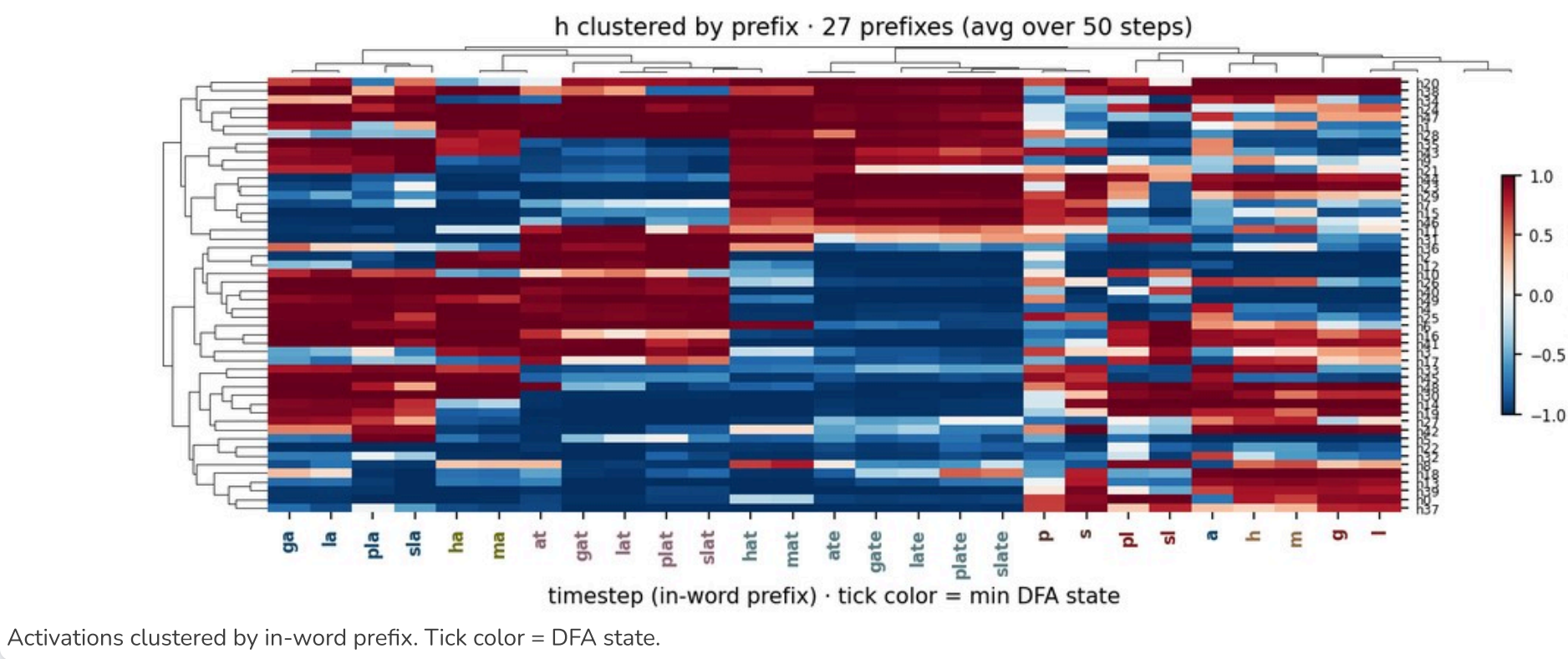
Softmax  $P(\text{next char} | \text{input so far})$ . Red dots = actual next character.

Fig 5 • Activation heatmap



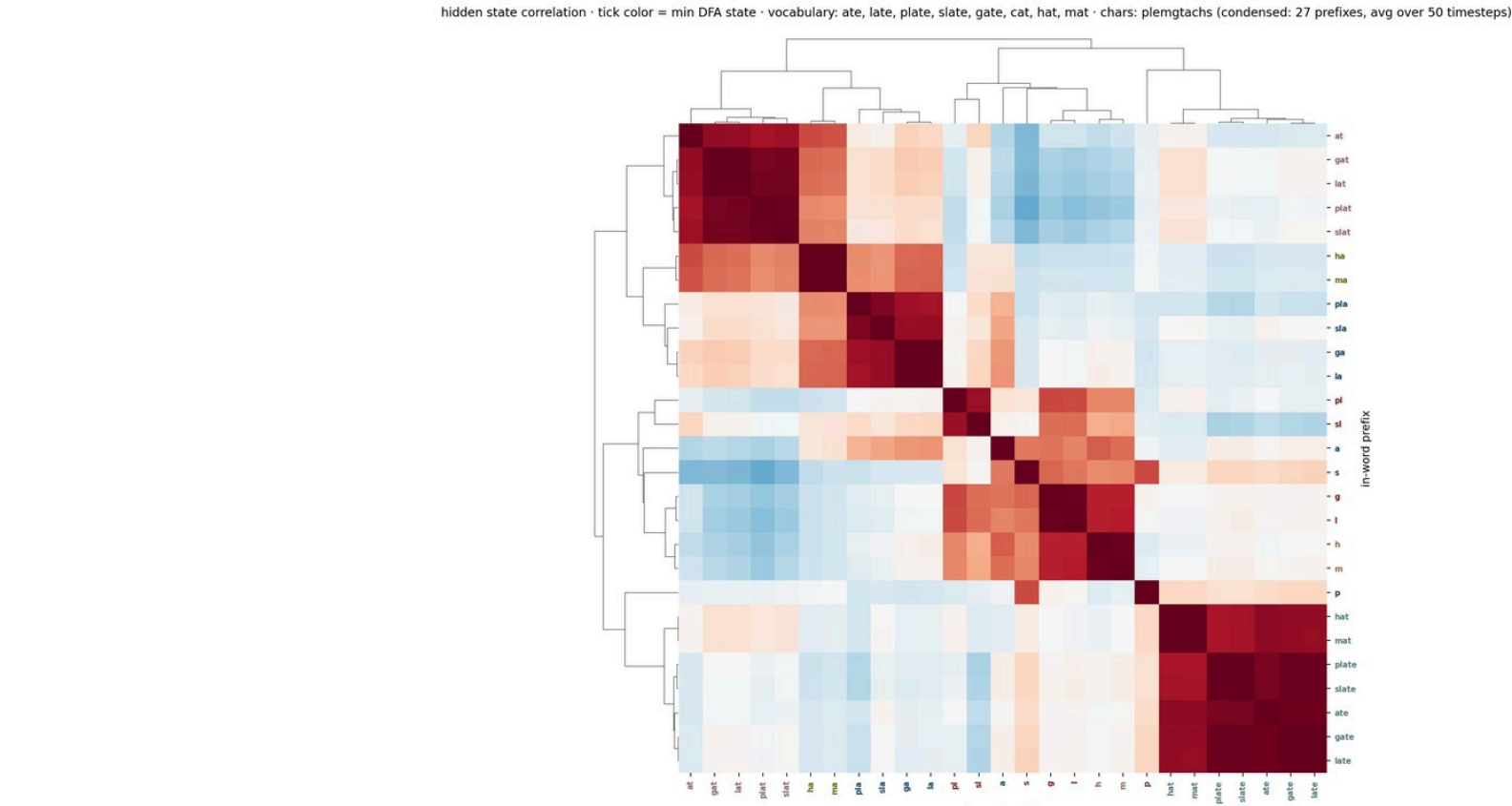
Hidden units over timesteps (input letters; units clustered).

Fig 6 • Activations by prefix



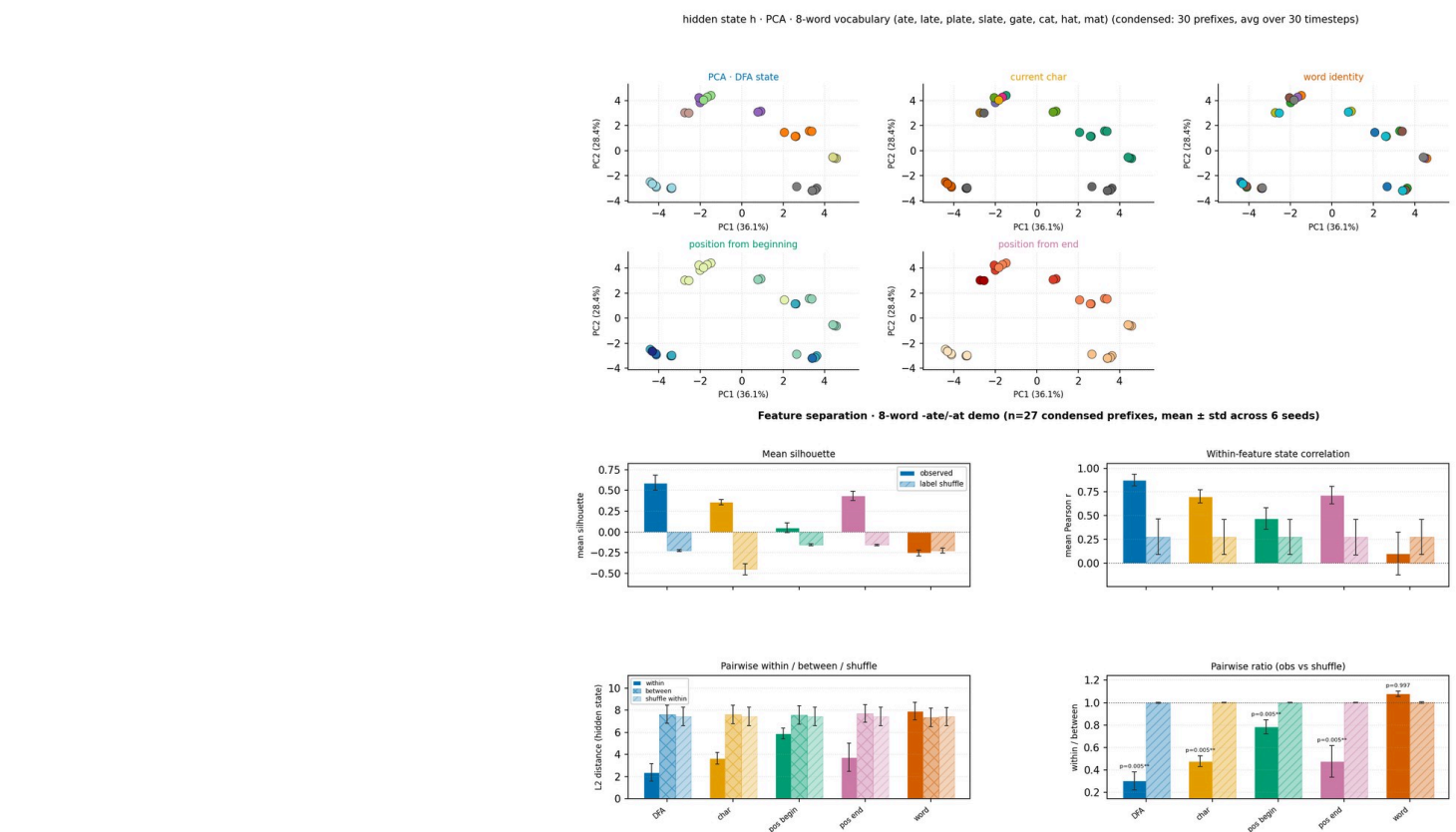
Activations clustered by in-word prefix. Tick color = min DFA state.

Fig 7 • Hidden-state correlation



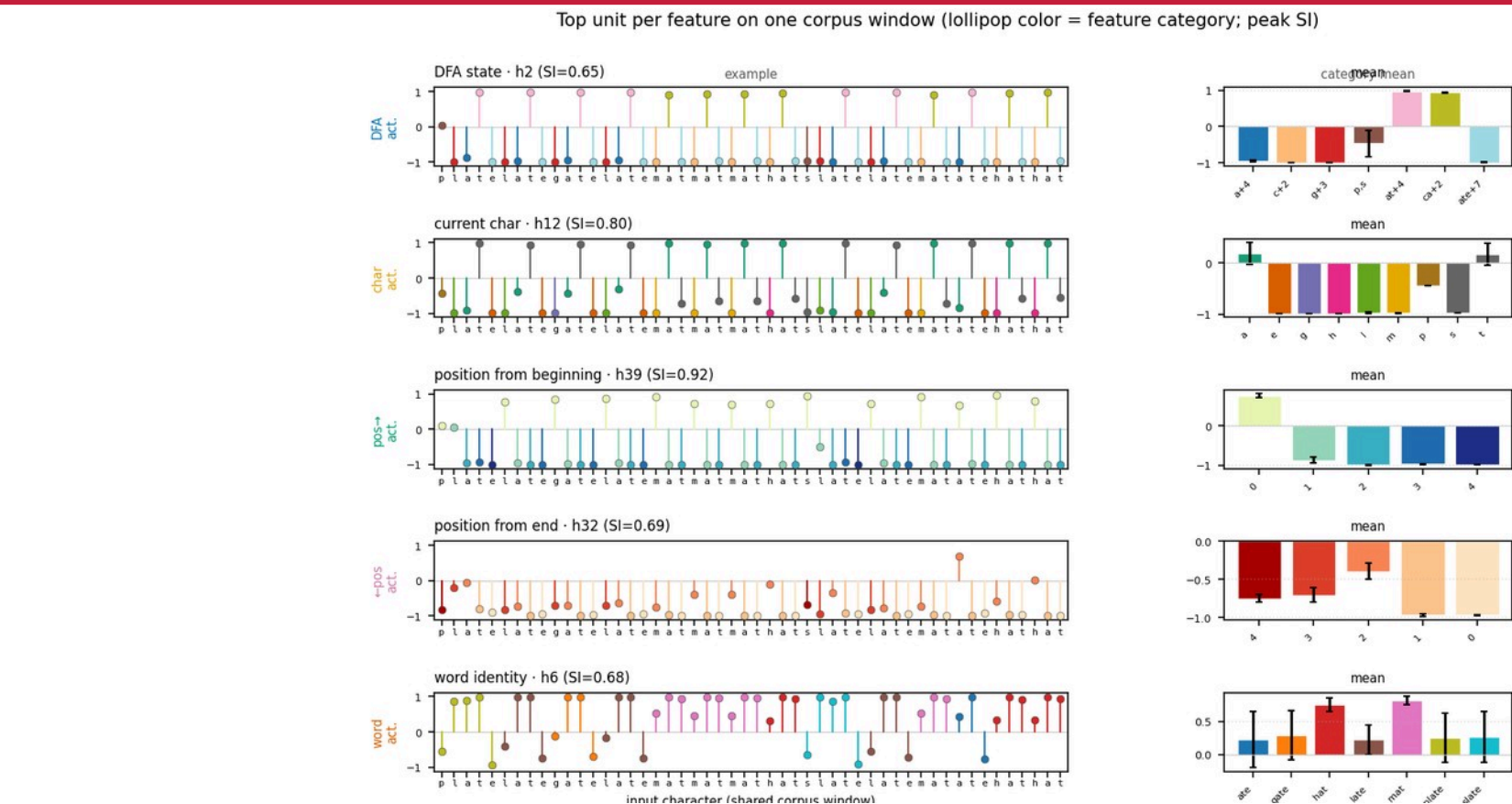
Prefix  $\times$  prefix correlation. Blocks merge prefixes that share a future.

Fig 8 • DFA geometry & separation



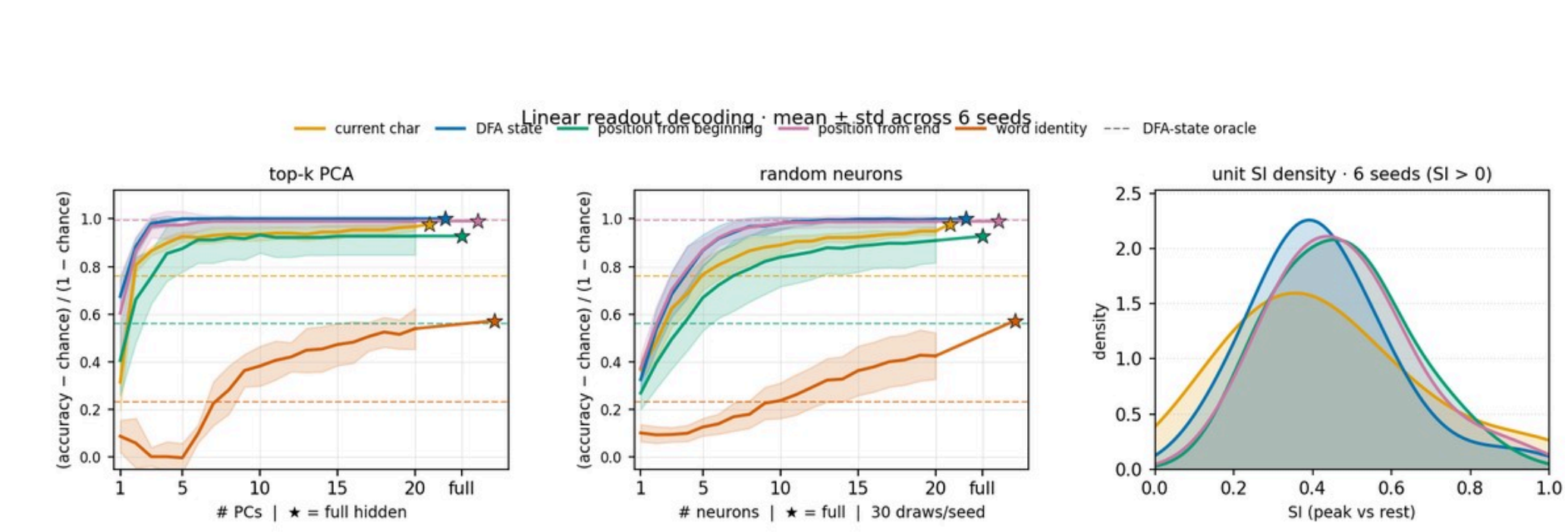
PCA of  $h$  by feature. DFA  $\eta^2 \approx 0.88$ ; word identity  $\approx 0.06$ .

Fig 9 • Example selective units



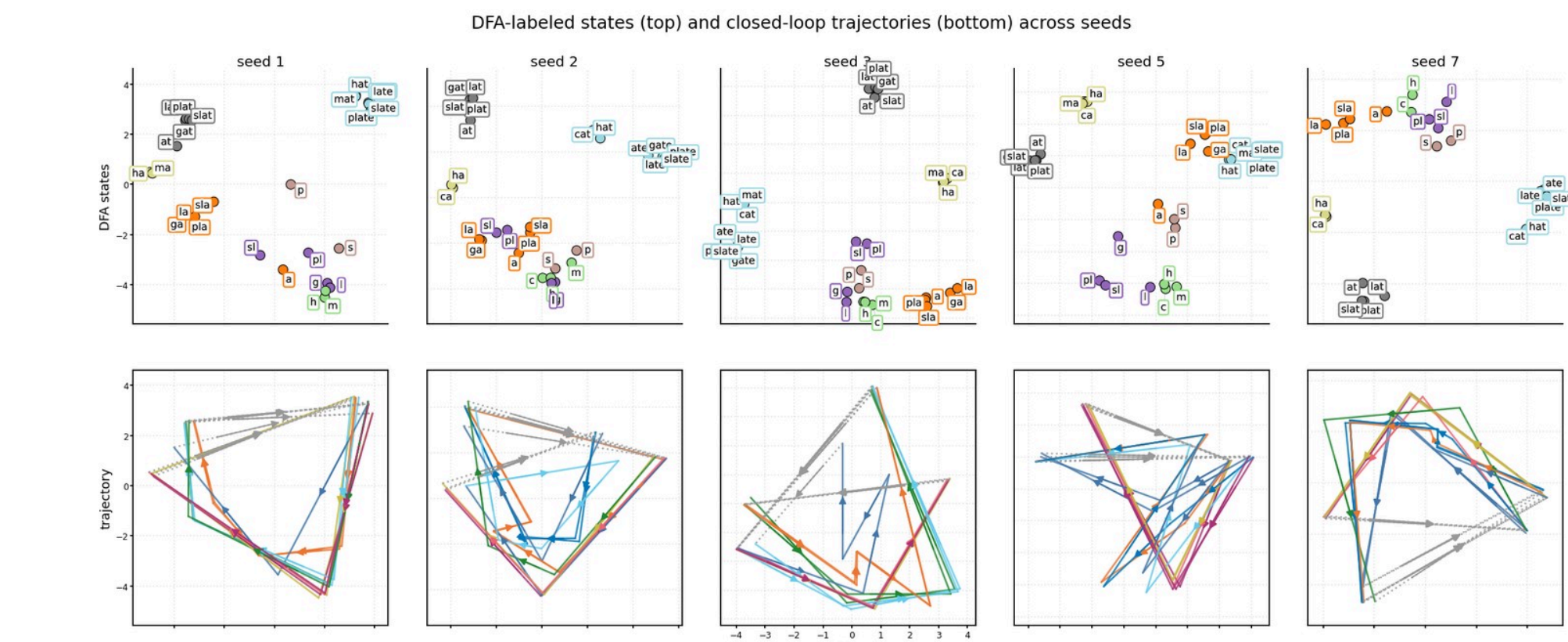
One top unit per feature on a shared corpus window.

Fig 10 • Linear decoding



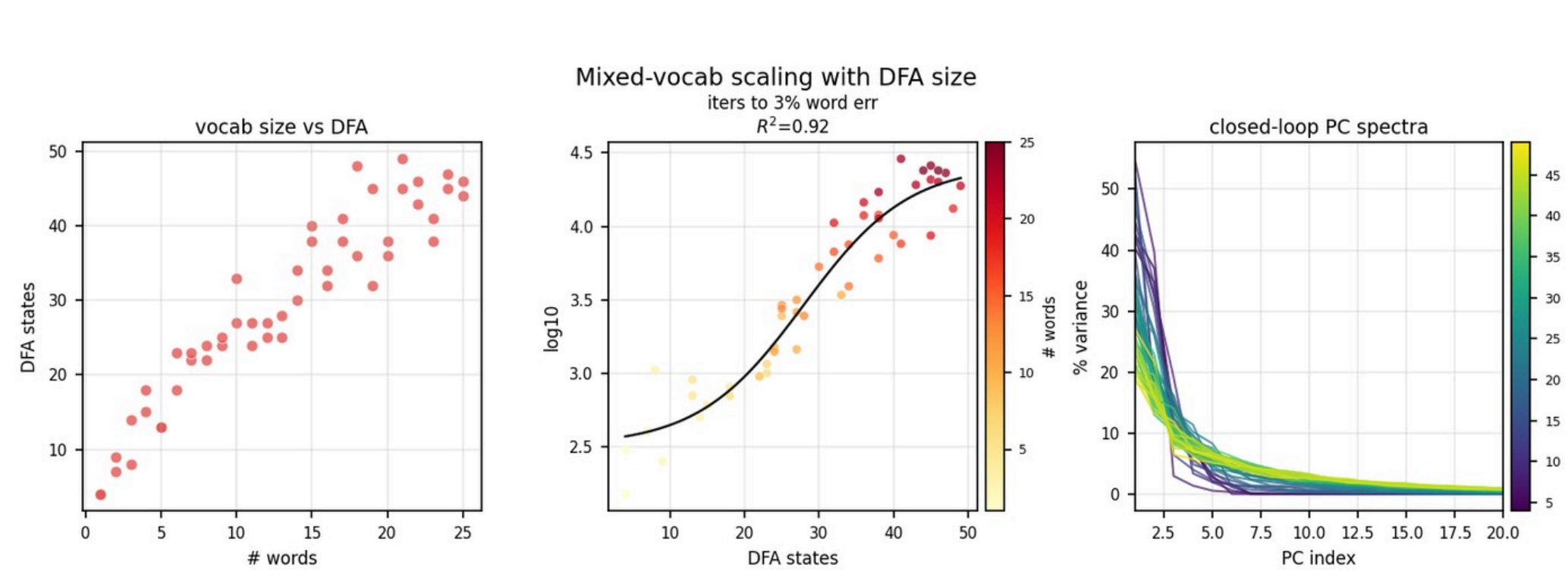
Mean  $\pm$  std, 6 seeds. DFA/character saturate in a few PCs; word identity needs many more.

Fig 11 • Word trajectories



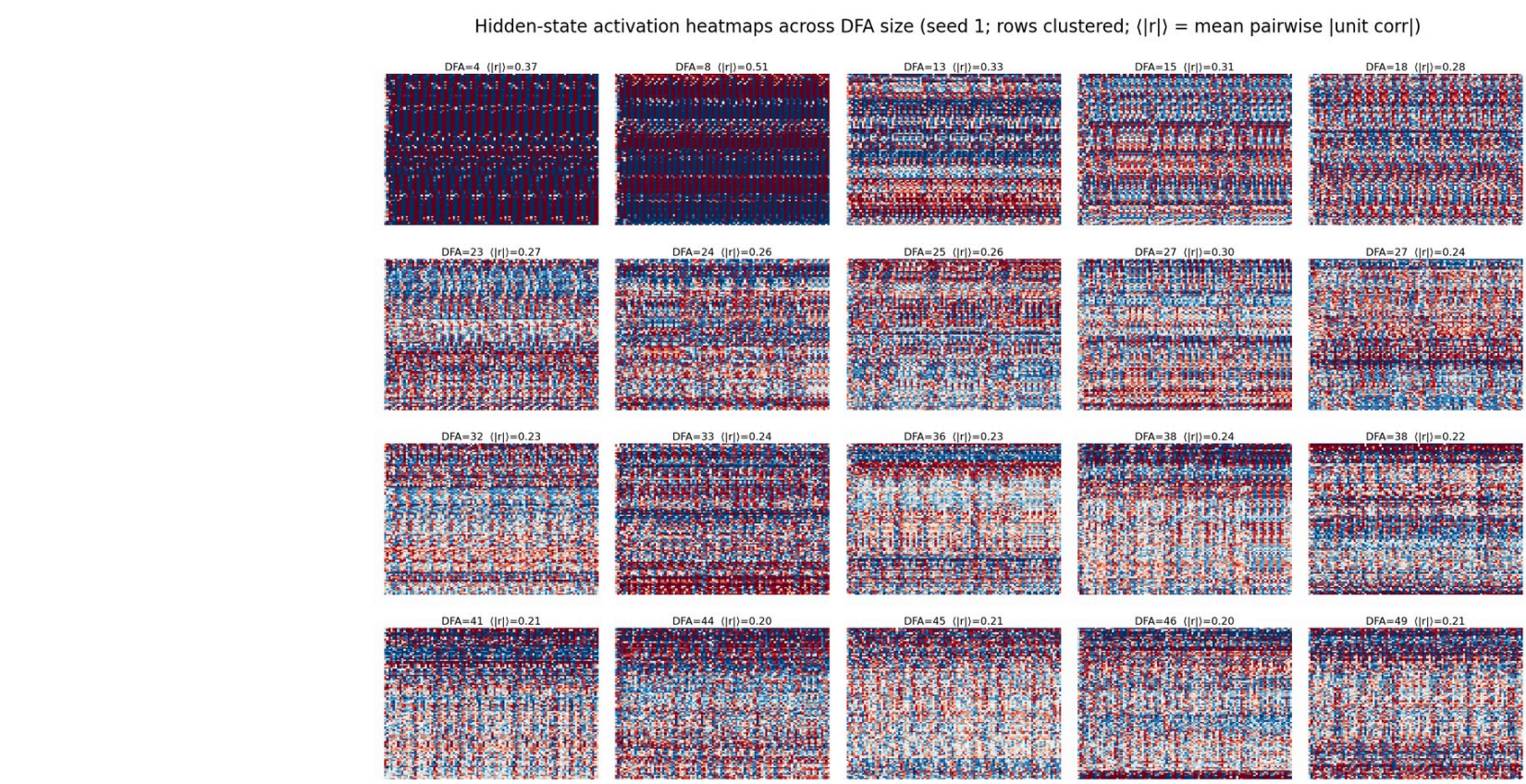
DFA-labeled PCA (top) and closed-loop trajectories (bottom) across five seeds.

Fig 12 • Mixed vocabacs track DFA size



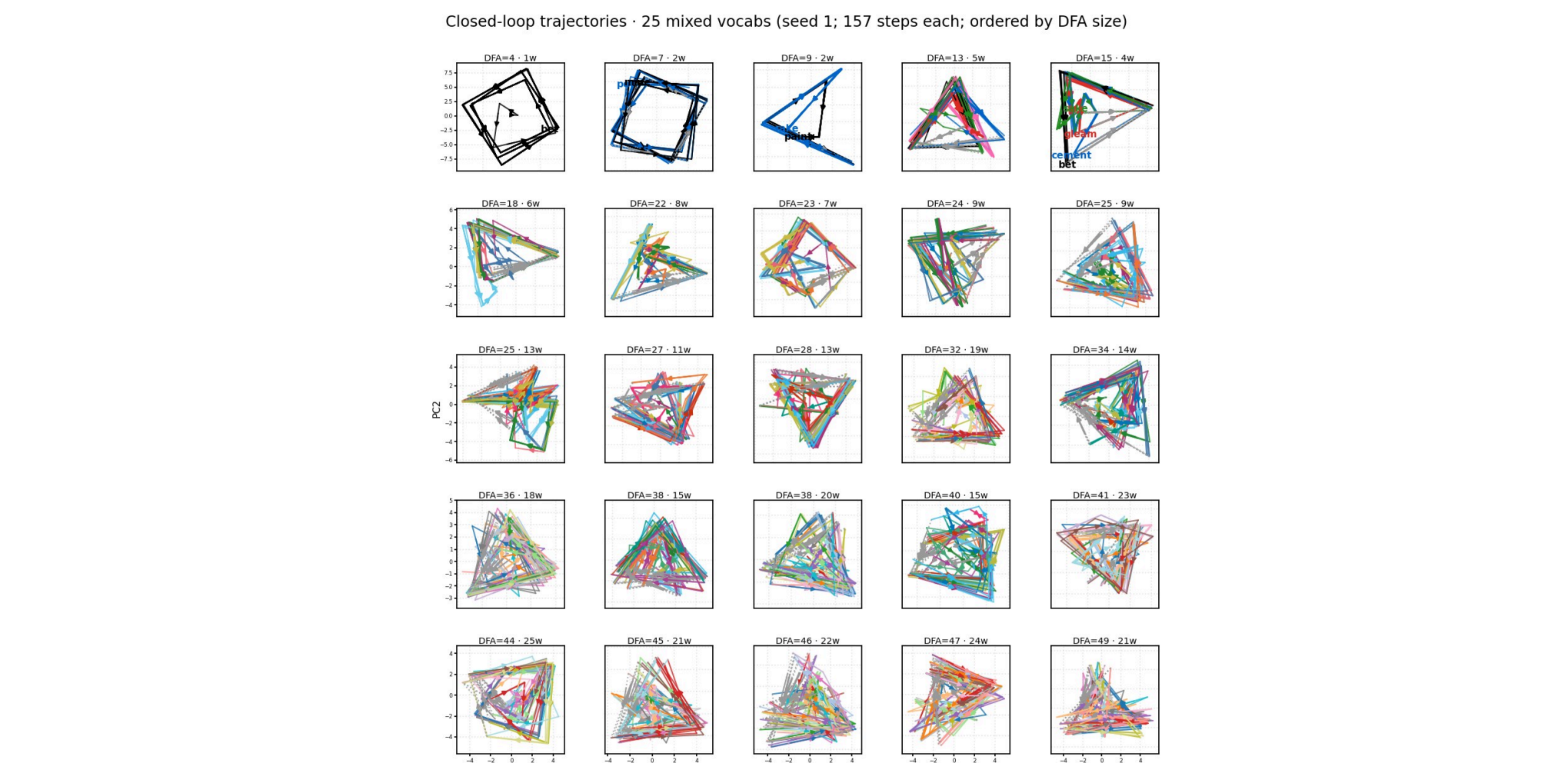
50 mixed English vocabacs ( $H=100$ ). Training cost and PC spectra track DFA size.

Fig 13 • Activations vs DFA size



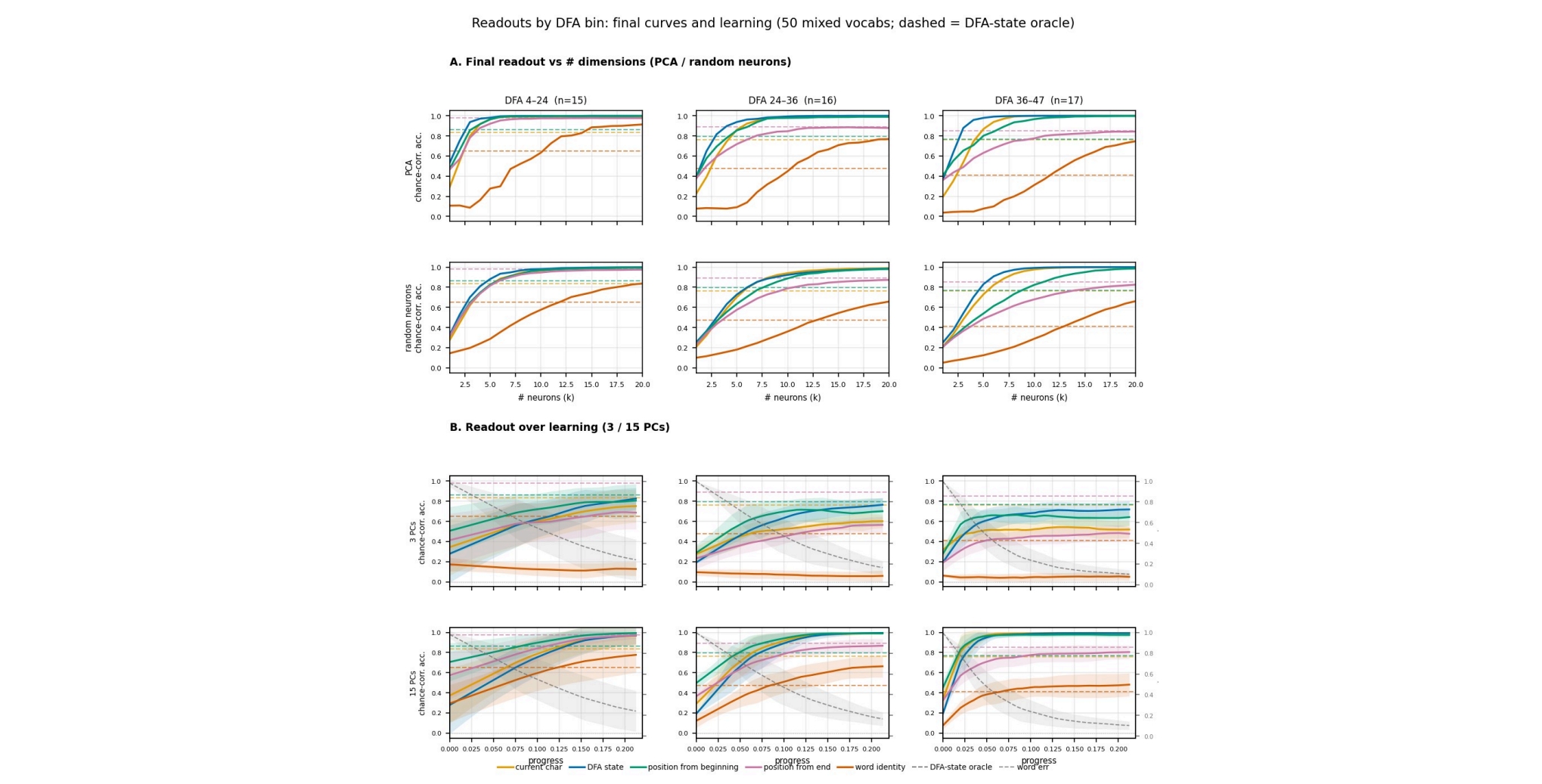
20 mixed vocabacs. Low-DFA: coarse bands; larger DFAs: denser, less repetitive raters.

Fig 14 • Trajectories vs DFA size



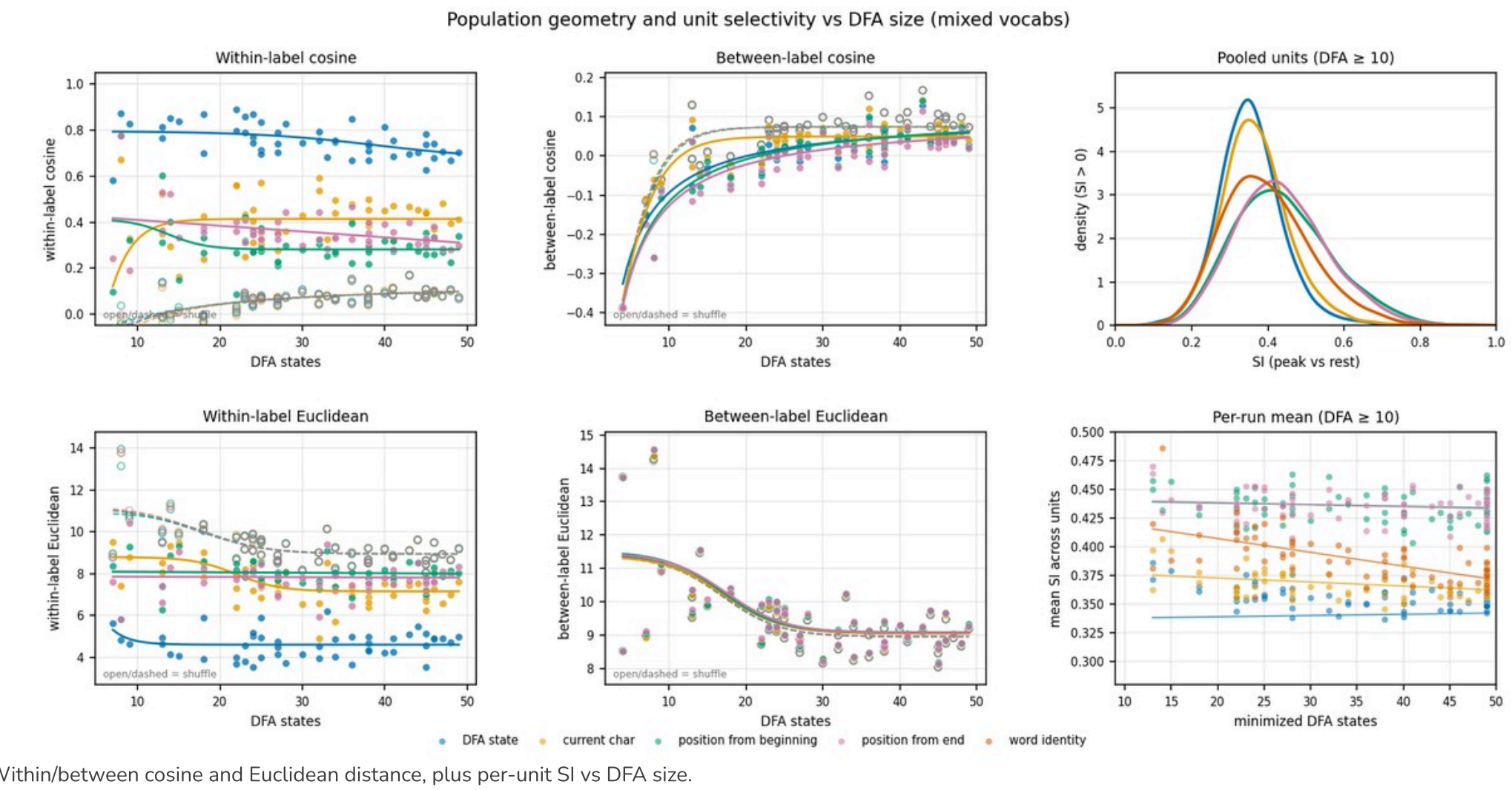
25 mixed vocabacs. Small automata: sparse motifs; larger DFAs crowd the PC plane.

Fig 15 • Readouts vs DFA size



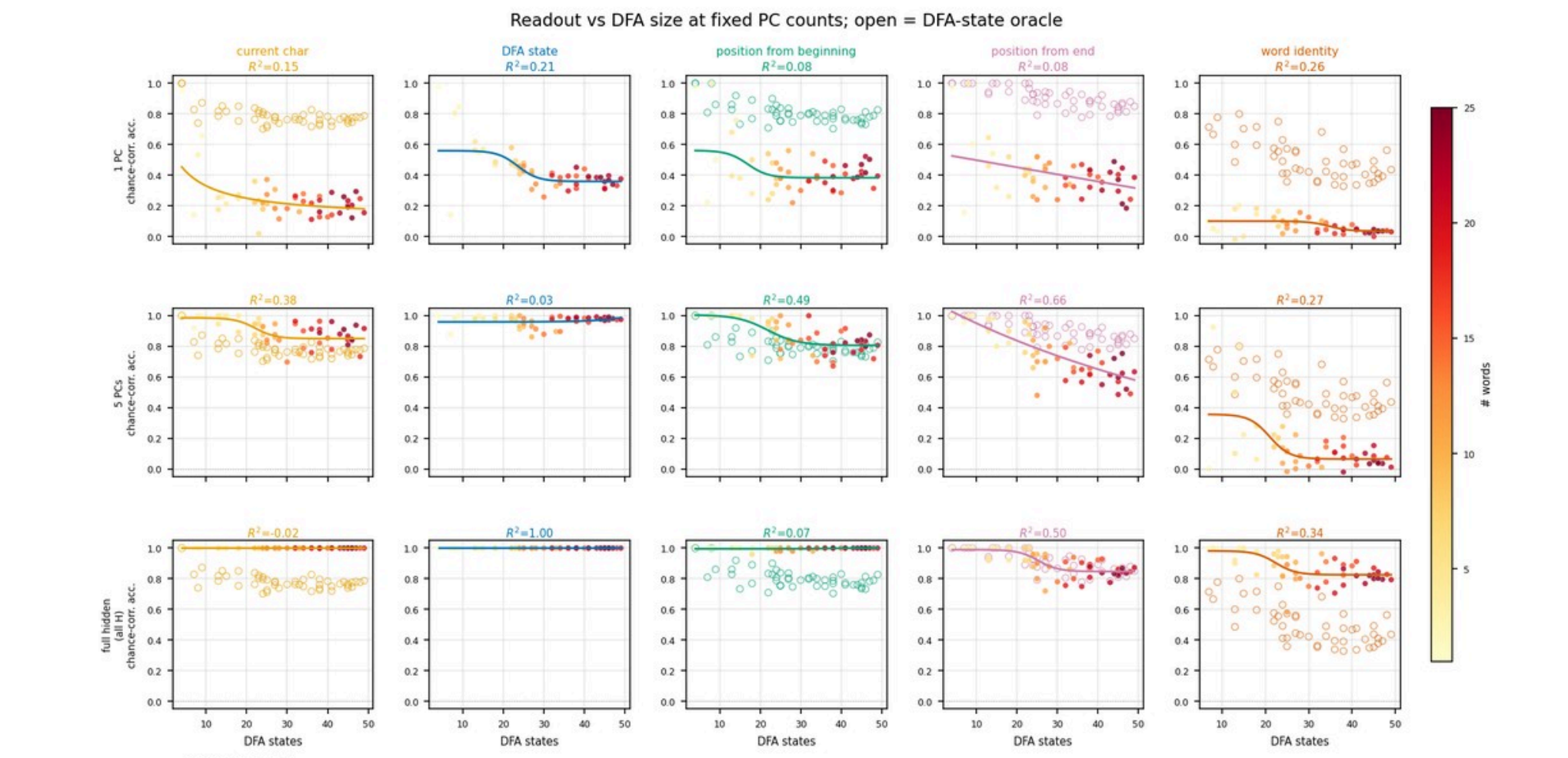
Chance-corrected readout by DFA-size bin. Word identity needs ~15 PCs.

Fig 16 • Geometry & unit selectivity



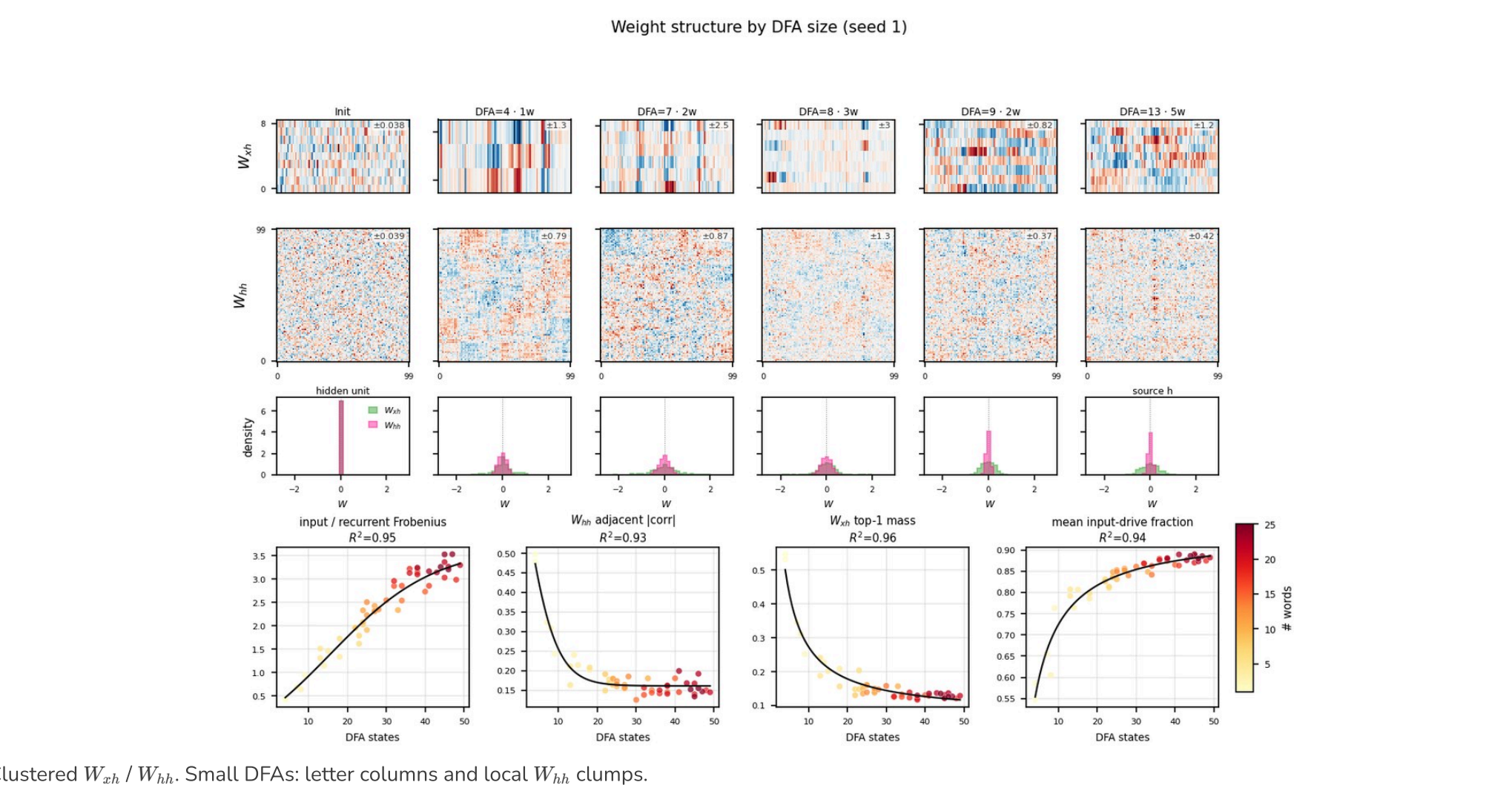
Within/between cosine and Euclidean distance, plus per-unit SI vs DFA size.

Fig 17 • Readout vs DFA at fixed PCs



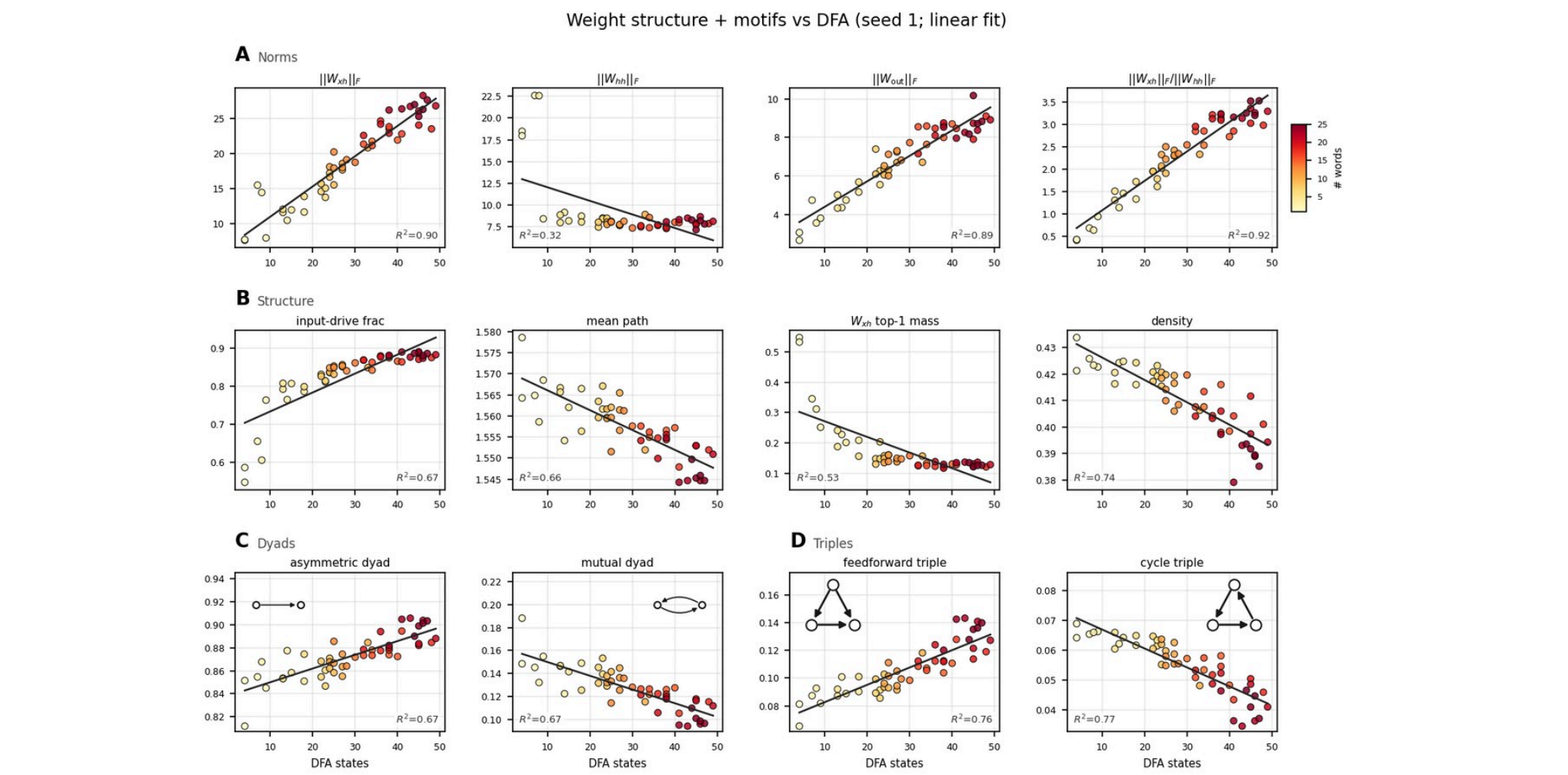
Accuracy vs DFA size at 1 PC, 5 PCs, and full  $H$ . Word identity falls off with DFA size.

Fig 18 • Weight matrices vs DFA



Clustered  $W_{ah}$  /  $W_{hh}$ . Small DFAs: letter columns and local  $W_{hh}$  clumps.

Fig 19 • Weight graph motifs vs DFA



Norms, structure, pairwise and 3-node motifs. Larger DFAs shift feedforward.